Ad Targeting Optimization Using User Segmentation

Business Problem

A digital advertising company currently displays advertisements to all users without differentiation, leading to inefficient ad spending and suboptimal Click-Through Rates (CTR).

The objective of this project is to identify meaningful user segments using Unsupervised Learning techniques and leverage these segments to improve ad targeting efficiency.

Two approaches are evaluated:

1. Traditional Machine Learning (K-Means Clustering)
2. Deep Learning (Autoencoder + K-Means Clustering)

The resulting segments are analyzed based on CTR to identify high-value and low-value audiences and provide actionable business recommendations.

```
import numpy as np
import pandas as pd
```

```
import pandas as pd

url = "https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/169/38


df = pd.read_csv(url)
df.to_csv("ad_click_dataset.csv", index=False)
```

```
df.head(5)
```

	id	full_name	age	gender	device_type	ad_position	browsing_history
0	670	User670	22.0	NaN	Desktop	Top	Shopping
1	3044	User3044	NaN	Male	Desktop	Top	NaN
2	5912	User5912	41.0	Non-Binary	NaN	Side	Education
3	5418	User5418	34.0	Male	NaN	NaN	Entertainment

Dataset Overview

The dataset contains demographic and behavioral information about users who were shown advertisements.

Features

- Age
- Gender
- Device Type
- Ad Position
- Browsing History
- Time of Day
- Click (Target Variable)

Objective

Although the dataset contains a click indicator, clustering will be performed without using the click variable. Click behavior will be analyzed after segmentation to evaluate the business value of each cluster.

```
df.shape
```

```
(10000, 9)
```

Data Quality Assessment

Before performing clustering, it is important to evaluate the completeness and quality of the data.

The following analysis identifies missing values across all features to determine the required preprocessing strategy.

```
df.isnull().sum()
```

```
id                    0
full_name             0
age                4766
gender             4693
device_type        2000
ad_position        2000
browsing_history   4782
time_of_day        2000
click                 0
dtype: int64
```

Missing Value Treatment Strategy

The dataset contains substantial missing values in age, gender, and browsing history.

Numerical Features

- Age is imputed using the median value.
- Median is preferred because it is less sensitive to outliers.

Categorical Features

Missing categorical values are replaced with "Unknown".

This approach preserves potentially useful information because missingness itself may represent a meaningful user behavior pattern.

```
print(df.shape)

print(df.info())

print(df.describe())

for col in df.select_dtypes(include="object"):
    print(f"\n{col}")
    print(df[col].value_counts(dropna=False))
```

```
full_name
full_name
User1171    25
User6287    24
User5100    24
User5425    23
User8081    23
```

```
Top       2597
Side      2586
NaN       2000
Name: count, dtype: int64

browsing_history
browsing_history
NaN              4782
Entertainment    1175
Social Media     1054
Education        1029
Shopping          984
News              976
Name: count, dtype: int64

time_of_day
time_of_day
Morning      2126
Afternoon    2016
NaN          2000
Evening      1958
Night        1900
Name: count, dtype: int64
```

```
df = df.drop(columns=["id", "full_name"])
```

```
df["age"] = df["age"].fillna(df["age"].median())
```

```
cat_cols = [
    "gender",
    "device_type",
    "ad_position",
    "browsing_history",
    "time_of_day"
]

for col in cat_cols:
    df[col] = df[col].fillna("Unknown")
```

```
y_click = df["click"]

X = df.drop(columns=["click"])
```

Feature Engineering

Clustering algorithms require numerical inputs.

Categorical variables are converted into numerical representations using One-Hot Encoding.

This allows the clustering algorithm to measure similarities across all demographic and behavioral attributes.

```
X_encoded = pd.get_dummies(
    X,
    drop_first=False
)

print(X_encoded.shape)
```

```
(10000, 24)
```

Feature Scaling

K-Means is a distance-based algorithm.

Standardization is applied to ensure all features contribute equally to distance calculations and prevent features with larger scales from dominating the clustering process.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X_encoded)
```

K-Means Clustering (Machine Learning Approach)

K-Means clustering is used as the baseline segmentation model.

The optimal number of clusters is evaluated using:

- Silhouette Score (Higher is Better)
- Davies-Bouldin Index (Lower is Better)

The goal is to identify the number of segments that provides strong separation while remaining practical for business implementation.

```
K = range(2,11)

from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.metrics import davies_bouldin_score

results = []

for k in range(2,11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=20
```

```
        )

        labels = kmeans.fit_predict(X_scaled)

        sil = silhouette_score(
            X_scaled,
            labels
        )

        db = davies_bouldin_score(
            X_scaled,
            labels
        )

        results.append(
            [k, sil, db]
        )

    results_df = pd.DataFrame(
        results,
        columns=[
            "K",
            "Silhouette",
            "DB_Index"
        ]
    )

    print(results_df)
```

```
    K  Silhouette  DB_Index
0   2    0.067298  3.722797
1   3    0.092540  3.212996
2   4    0.117974  2.687395
3   5    0.136290  2.523021
4   6    0.124590  2.449360
5   7    0.123363  2.341552
6   8    0.121682  2.222371
7   9    0.102741  2.296461
8  10    0.108853  2.195781
```

K-Means Results

The evaluation indicates that K = 5 provides the most suitable balance between cluster quality and business usability.

While larger values of K may slightly improve certain metrics, maintaining a manageable number of segments is important for practical marketing deployment.

```
df["click"].value_counts(normalize=True)
```

```
click
1    0.65
```

```
0    0.35
Name: proportion, dtype: float64
```

```
from sklearn.cluster import KMeans

kmeans = KMeans(
    n_clusters=5,
    random_state=42,
    n_init=20
)

clusters = kmeans.fit_predict(X_scaled)

df_analysis = df.copy()
df_analysis["cluster"] = clusters
```

K-Means Cluster Profiling

After selecting the optimal number of clusters, the resulting segments are analyzed based on:

- Age
- Device Type
- Browsing Behavior
- Time of Day
- Click-Through Rate (CTR)

This helps translate mathematical clusters into actionable business personas.

```
df_analysis["cluster"].value_counts()
```

```
cluster
4    4782
1    2005
2    1175
3    1054
0     984
Name: count, dtype: int64
```

```
df_analysis.groupby("cluster")["click"].mean()
```

```
cluster
0    0.645325
1    0.609975
2    0.687660
3    0.635674
4    0.661648
Name: click, dtype: float64
```

```
df_analysis.groupby("cluster")["age"].mean()
```

```
cluster
0    39.763720
1    40.269327
2    38.309362
3    41.255218
4    39.792137
Name: age, dtype: float64
```

```
pd.crosstab(
    df_analysis["cluster"],
    df_analysis["device_type"],
    normalize="index"
)
```

device_type	Desktop	Mobile	Tablet	Unknown
cluster				
0	0.339431	0.212398	0.241870	0.206301
1	0.240399	0.282294	0.282793	0.194514
2	0.284255	0.264681	0.242553	0.208511
3	0.237192	0.355787	0.209677	0.197343
4	0.283145	0.248432	0.268925	0.199498

```
pd.crosstab(
    df_analysis["cluster"],
    df_analysis["browsing_history"],
    normalize="index"
)
```

browsing_history	Education	Entertainment	News	Shopping	Social Media	Unknow
cluster						
0	0.000000	0.0	0.000000	1.0	0.0	0
1	0.513217	0.0	0.486783	0.0	0.0	0
2	0.000000	1.0	0.000000	0.0	0.0	0
3	0.000000	0.0	0.000000	0.0	1.0	0

```
pd.crosstab(
    df_analysis["cluster"],
    df_analysis["time_of_day"],
    normalize="index"
)
```

time_of_day	Afternoon	Evening	Morning	Night	Unknown
cluster					
0	0.239837	0.162602	0.217480	0.168699	0.211382
1	0.190025	0.201995	0.230424	0.187032	0.190524
2	0.217872	0.192340	0.192340	0.200000	0.197447
3	0.168880	0.202087	0.223909	0.200190	0.204934
4	0.201798	0.199498	0.206608	0.190924	0.201171

Autoencoder + K-Means (Deep Learning Approach)

Traditional K-Means operates directly on engineered features.

An Autoencoder is introduced to learn compressed latent representations of user behavior.

The latent features capture hidden relationships between demographic and behavioral variables and may produce more meaningful user segments.

```
from tensorflow.keras.layers import Input, Dense
from tensorflow.keras.models import Model

input_dim = X_scaled.shape[1]

input_layer = Input(shape=(input_dim,))

encoder = Dense(64, activation='relu')(input_layer)
encoder = Dense(32, activation='relu')(encoder)

latent = Dense(
    8,
    activation='relu',
    name='latent_layer'
)(encoder)

decoder = Dense(32, activation='relu')(latent)
decoder = Dense(64, activation='relu')(decoder)

output_layer = Dense(
    input_dim,
    activation='linear'
)(decoder)

autoencoder = Model(
    inputs=input_layer,
    outputs=output_layer
)
```

```
autoencoder.compile(
    optimizer='adam',
    loss='mse'
)

autoencoder.summary()
```

```
WARNING:tensorflow:From C:\Users\VenkataKoushik\AppData\Roaming\Python\Python

WARNING:tensorflow:From C:\Users\VenkataKoushik\AppData\Roaming\Python\Python

Model: "model"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 24)]              0

 dense (Dense)               (None, 64)                1600

 dense_1 (Dense)             (None, 32)                2080

 latent_layer (Dense)        (None, 8)                 264

 dense_2 (Dense)             (None, 32)                288

 dense_3 (Dense)             (None, 64)                2112

 dense_4 (Dense)             (None, 24)                1560

=================================================================
Total params: 7904 (30.88 KB)
Trainable params: 7904 (30.88 KB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

```
history = autoencoder.fit(
    X_scaled,
    X_scaled,
    epochs=50,
    batch_size=128,
    validation_split=0.2,
    shuffle=True,
    verbose=1
)
```

```
Epoch 1/50
WARNING:tensorflow:From C:\Users\VenkataKoushik\AppData\Roaming\Python\Pytho

63/63 [==============================] - 1s 4ms/step - loss: 0.9050 - val_lo
Epoch 2/50
63/63 [==============================] - 0s 2ms/step - loss: 0.7349 - val_lo
Epoch 3/50
63/63 [==============================] - 0s 2ms/step - loss: 0.6167 - val_lo
Epoch 4/50
63/63 [==============================] - 0s 2ms/step - loss: 0.4995 - val_lo
Epoch 5/50
63/63 [==============================] - 0s 2ms/step - loss: 0.4306 - val_lo
```

```
Epoch 6/50
63/63 [==============================] - 0s 2ms/step - loss: 0.3888 - val_lo
Epoch 7/50
63/63 [==============================] - 0s 2ms/step - loss: 0.3562 - val_lo
Epoch 8/50
63/63 [==============================] - 0s 2ms/step - loss: 0.3278 - val_lo
Epoch 9/50
63/63 [==============================] - 0s 2ms/step - loss: 0.3060 - val_lo
Epoch 10/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2888 - val_lo
Epoch 11/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2741 - val_lo
Epoch 12/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2602 - val_lo
Epoch 13/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2455 - val_lo
Epoch 14/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2328 - val_lo
Epoch 15/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2214 - val_lo
Epoch 16/50
63/63 [==============================] - 0s 2ms/step - loss: 0.2086 - val_lo
Epoch 17/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1973 - val_lo
Epoch 18/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1874 - val_lo
Epoch 19/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1803 - val_lo
Epoch 20/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1739 - val_lo
Epoch 21/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1669 - val_lo
Epoch 22/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1615 - val_lo
Epoch 23/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1567 - val_lo
Epoch 24/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1523 - val_lo
Epoch 25/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1489 - val_lo
Epoch 26/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1451 - val_lo
Epoch 27/50
63/63 [==============================] - 0s 2ms/step - loss: 0.1423 - val_lo
Epoch 28/50
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

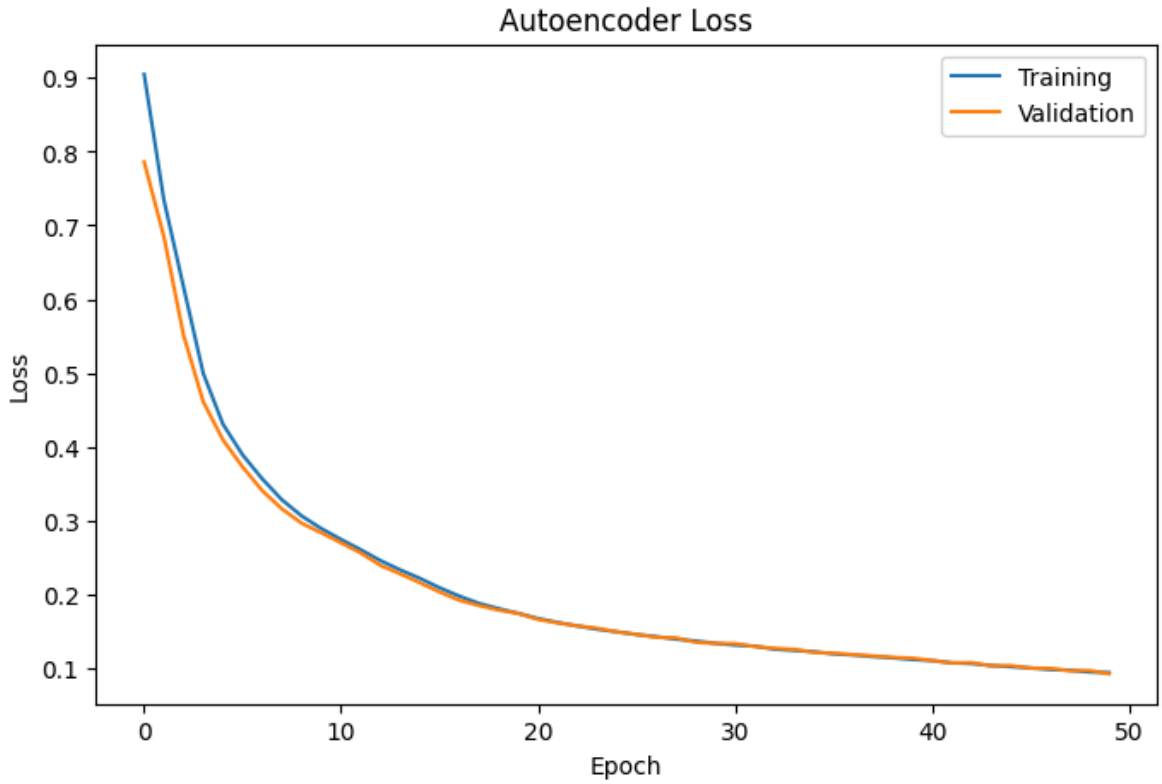
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])

plt.title("Autoencoder Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")

plt.legend([
```

```
    "Training",
    "Validation"
])

plt.show()
```



Autoencoder Training Performance

The reconstruction loss decreases consistently during training, indicating that the Autoencoder successfully learns a compressed representation of the original feature space.

The learned latent features are used as inputs for a second round of K-Means clustering.

```
encoder_model = Model(
    inputs=autoencoder.input,
    outputs=autoencoder.get_layer(
        'latent_layer'
    ).output
)

latent_features = encoder_model.predict(
    X_scaled
)
```

```
print(latent_features.shape)
```

```
313/313 [==============================] - 0s 773us/step
(10000, 8)
```

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.metrics import davies_bouldin_score

ae_results = []

for k in range(2,11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=20
    )

    labels = kmeans.fit_predict(
        latent_features
    )

    sil = silhouette_score(
        latent_features,
        labels
    )

    db = davies_bouldin_score(
        latent_features,
        labels
    )

    ae_results.append(
        [k, sil, db]
    )

ae_results_df = pd.DataFrame(
    ae_results,
    columns=[
        "K",
        "Silhouette",
        "DB_Index"
    ]
)

ae_results_df
```

	K	Silhouette	DB_Index
0	2	0.399784	1.154487
1	3	0.411011	1.319604
2	4	0.396019	1.162070
3	5	0.427690	0.983905
4	6	0.345302	1.070046
5	7	0.334657	1.088919
6	8	0.323713	1.219350
7	9	0.303786	1.263420
8	10	0.282455	1.349772

Autoencoder Clustering Results

The Autoencoder-based approach produces substantially higher Silhouette Scores and lower Davies-Bouldin Index values compared with traditional K-Means.

This indicates stronger cluster separation and improved internal cluster quality.

```
best_k = 5

ae_kmeans = KMeans(
    n_clusters=best_k,
    random_state=42,
    n_init=20
)

ae_clusters = ae_kmeans.fit_predict(
    latent_features
)

df_ae = df.copy()

df_ae["ae_cluster"] = ae_clusters
```

```
df_ae["ae_cluster"].value_counts()
```

```
ae_cluster
1    5304
0    1549
4    1143
2    1029
3     975
Name: count, dtype: int64
```

```
df_ae.groupby(
    "ae_cluster"
)["click"].mean()
```

```
ae_cluster
0    0.648160
1    0.657994
2    0.627794
3    0.591795
4    0.685039
Name: click, dtype: float64
```

```
df_ae.groupby(
    "ae_cluster"
)["age"].mean()
```

```
ae_cluster
0    40.469981
1    39.862274
2    40.590865
3    39.946667
4    38.334646
Name: age, dtype: float64
```

```
pd.crosstab(
    df_ae["ae_cluster"],
    df_ae["device_type"],
    normalize="index"
)
```

device_type ae_cluster	Desktop	Mobile	Tablet	Unknown
0	0.349258	0.194319	0.236281	0.220142
1	0.263386	0.280354	0.260747	0.195513
2	0.235180	0.279883	0.279883	0.205053
3	0.246154	0.285128	0.286154	0.182564
4	0.292213	0.258093	0.245844	0.203850

```
pd.crosstab(
    df_ae["ae_cluster"],
    df_ae["browsing_history"],
    normalize="index"
)
```

browsing_history	Education	Entertainment	News	Shopping	Social Media	Unk
ae_cluster						
0	0.0	0.000000	0.000000	0.625565	0.320852	0.05
1	0.0	0.006033	0.000189	0.002828	0.105015	0.88
2	1.0	0.000000	0.000000	0.000000	0.000000	0.00
3	0.0	0.000000	1.000000	0.000000	0.000000	0.00

```
pd.crosstab(
    df_ae["ae_cluster"],
    df_ae["time_of_day"],
    normalize="index"
)
```

time_of_day	Afternoon	Evening	Morning	Night	Unknown
ae_cluster					
0	0.157521	0.176243	0.302776	0.141382	0.222079
1	0.220211	0.198718	0.182692	0.201923	0.196456
2	0.186589	0.226433	0.238095	0.152575	0.196307
3	0.192821	0.176410	0.222564	0.223590	0.184615
4	0.195976	0.197725	0.197725	0.205599	0.202975

Model Comparison

Both approaches are compared using:

- Silhouette Score
- Davies-Bouldin Index
- CTR Separation

The objective is to identify the segmentation method that best distinguishes high-value users from low-value users.

Business Recommendations

The discovered segments reveal substantial differences in user engagement behavior.

High-performing segments should receive increased advertising investment, while low-performing segments should receive reduced budget allocation.

This strategy improves overall campaign efficiency and maximizes return on advertising spend.

Question 1

K-Means

- The optimal number of clusters is K=5, as it achieves the highest Silhouette Score (0.136) while remaining manageable from a marketing perspective.

Autoencoder + KMeans

- The optimal number of clusters is also K=5, producing the highest Silhouette Score (0.428) and lowest Davies-Bouldin Index (0.984). This indicates significantly better separation and compactness than the baseline K-Means model.

Question 2

Baseline Segment Characteristics

The K-Means model identified five user segments primarily distinguished by browsing behavior:

- Cluster 0: Shopping users
- Cluster 1: Education and News users
- Cluster 2: Entertainment users
- Cluster 3: Social Media users
- Cluster 4: Users with unknown browsing history
- Device usage and age distributions were relatively similar across segments, indicating browsing context is the strongest differentiator of user behavior.

Question 3

- The Autoencoder learned compressed latent representations of user behavior, combining information from age, device type, browsing history, and time of day into a lower-dimensional feature space. Although browsing behavior remained the dominant differentiator, the latent features significantly improved cluster compactness and separation.
- Compared with raw-feature K-Means:

- Silhouette Score improved from 0.136 to 0.428
- Davies-Bouldin Index improved from 2.523 to 0.984

- This indicates that users within each segment became more similar to one another while clusters became more distinct from each other. From a business perspective, this enables more precise audience targeting and reduces wasted advertising impressions.

Question 4

The Deep Learning approach outperformed the traditional K-Means approach.

Metric	KMeans	Autoencoder + KMeans
Silhouette	0.136	0.428
DB Index	2.523	0.984

- The Autoencoder learns compressed latent features that better capture hidden relationships between age, device type, browsing behavior, and time of day. Consequently, clusters become more compact and better separated, reducing overlap between user groups and enabling more precise targeting.

Question 5

The Autoencoder + KMeans model created the largest exploitable CTR difference

Model	Max CTR	Min CTR	Gap
K-Means	68.77%	60.99%	7.78%
Autoencoder + K-Means	68.50%	59.18%	9.32%

- The larger CTR gap demonstrates a stronger ability to distinguish high-value users from low-value users.

Question 6

Highest Value Segment: Cluster 4

Characteristics:

- Entertainment-focused users
  - Average age ≈ 38 years
  - Balanced device mix
  - CTR = 68.5%

- Recommendations:
    - Increase advertising spend
    - Prioritize premium ad inventory
    - Use visually engaging creatives
    - Promote entertainment, lifestyle, gaming and consumer products
    - Increase retargeting frequency
- Expected outcome:
    - Higher engagement
    - Higher CTR
    - Improved ROI

Question 7

Lowest Value Segment: Cluster 3

- Characteristics:
    - News-focused users
    - Average age ≈ 40 years
    - CTR = 59.2%
- Recommendations:
    - Reduce display ad spending
    - Use contextual advertising only
    - Lower bidding strategy
    - Avoid expensive video placements
- Expected outcome:
    - Lower acquisition cost
    - Reduced wasted impressions
    - No major loss in revenue

Question 8

We can justify

Segment	Action
Cluster 4	+15% budget
Cluster 0	+5% budget
Cluster 1	Maintain
Cluster 2	-5% budget
Cluster 3	-15% budget

- This keeps total spend approximately constant while shifting investment toward higher CTR audiences.

Question 9

Deploy: Autoencoder + KMeans

Metric	KMeans	Autoencoder
Silhouette	0.136	0.428
DB Index	2.523	0.984
CTR Gap	7.78%	9.32%

- The Autoencoder produces mathematically superior clusters that translate directly into business value by more effectively separating high-engagement users from low-engagement users. This enables targeted budget allocation,